🔑 **Answer Key — Teacher Copy · Lesson 2-2**

MSTAR PRACTICE WORKSHEET

Lesson 2-2: Represent and Describe Data in a Histogram

Grade 6 Mathematics · Standard 6.DS.5

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

A histogram shows these frequencies: 0–4: 3 players, 5–9: 8 players, 10–14: 12 players, 15–19: 5 players. How many players are represented in total?

A. 12

B. 15

C. 20

D. 28

Add all frequencies: 3 + 8 + 12 + 5 = 28 players.

Question 2 SELECTED RESPONSE — CORRECT: A

In a histogram, what does the height of each bar represent?

A. The frequency (count) of data in that interval

B. The average of the data in that interval

C. The range of the interval

D. The median of the data

Each bar's height shows the frequency — how many data values fall within that interval.

Question 3 SELECTED RESPONSE — CORRECT: A

How is a histogram different from a regular bar graph?

- (A) Histogram bars touch (no gaps) because the intervals are continuous ranges
- (B) There is no difference
- (C) Histograms use colors and bar graphs don't
- (D) Bar graphs show numbers and histograms show words

In a histogram, bars touch because each interval connects to the next (0–9, 10–19, etc.) — the data is continuous.

Question 4 SELECTED RESPONSE — CORRECT: B

A histogram has intervals: 0–4, 5–9, 10–14, 15–19. What is the width of each interval?

- (A) 4
- (B) 5
- (C) 10
- (D) 19

Each interval covers 5 values (0,1,2,3,4 then 5,6,7,8,9 etc.), so the width is 5.

Question 5 SELECTED RESPONSE — CORRECT: D

A histogram bar for the interval 20–29 reaches a height of 7. What does the 7 tell you?

- (A) The largest value is 7
- (B) There are 7 intervals
- (C) The interval is 7 wide
- (D) 7 data values fall between 20 and 29

A bar's height is the frequency — here, 7 values land in the 20–29 interval.

Question 6 SELECTED RESPONSE — CORRECT: B

A histogram of test scores shows: 50–59: 2, 60–69: 5, 70–79: 10, 80–89: 8, 90–99: 3. Which interval has the most students?

A. 60–69 with 5 students

B. 70–79 with 10 students

C. 80–89 with 8 students

D. 90–99 with 3 students

The 70–79 interval has the highest frequency of 10 students.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A histogram uses intervals 0–9, 10–19, 20–29, and 30–39. A data value of 19 belongs in which interval?

- (A) 10–19
- (B) 0–9
- (C) 20–29
- (D) It belongs in two intervals.

> 19 falls between 10 and 19 inclusive, so it belongs in the 10–19 bar. Intervals must not overlap, so every value lands in exactly one.

- **B:** 19 is larger than 9, so it cannot sit in that bar. Read the upper bound of each interval.
- **C:** That interval starts at 20, and 19 is below it. The 10-19 interval includes its endpoint.
- **D:** Histogram intervals are built so they never overlap, so every value lands in exactly one bar.

Part B — correct answer: B

Why are the bars of a histogram drawn with no gaps between them?

- (A) Gaps would make the histogram too wide.
- (B) The intervals are continuous, so the bars show one unbroken range of values.
- (C) Histograms always use exactly four bars.
- (D) The bars represent categories that must touch.

> A histogram displays numerical data over continuous intervals. Touching bars show that the number line runs without breaks. A bar graph of categories DOES have gaps.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about histograms:

- (A) The intervals must be equal in width.
- (B) The height of each bar is the frequency of values in that interval.
- (C) Bars touch because the intervals are continuous.
- (D) You can read every individual data value directly off a histogram.
- (E) Intervals must not overlap.

> A, B, C, and E are true. D is false — a histogram shows how many values fall in each interval, but the individual values are lost inside the bar.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A teacher collected test scores and made two different histograms — one with intervals of 5 (50–54, 55–59, etc.) and one with intervals of 20 (50–69, 70–89, 90–109). Both use the same data. How might the two histograms look different? Which interval size gives you more detail about the distribution? When might the larger interval be better?

Model answer: Intervals of 5 across 50–109 make 12 narrow bars with small frequencies, so peaks, gaps, and the exact shape show up clearly. Intervals of 20 make only 3 tall bars that smooth those details away, which is useful when the data set is large or you only want the big-picture comparison of low, middle, and high scores.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.